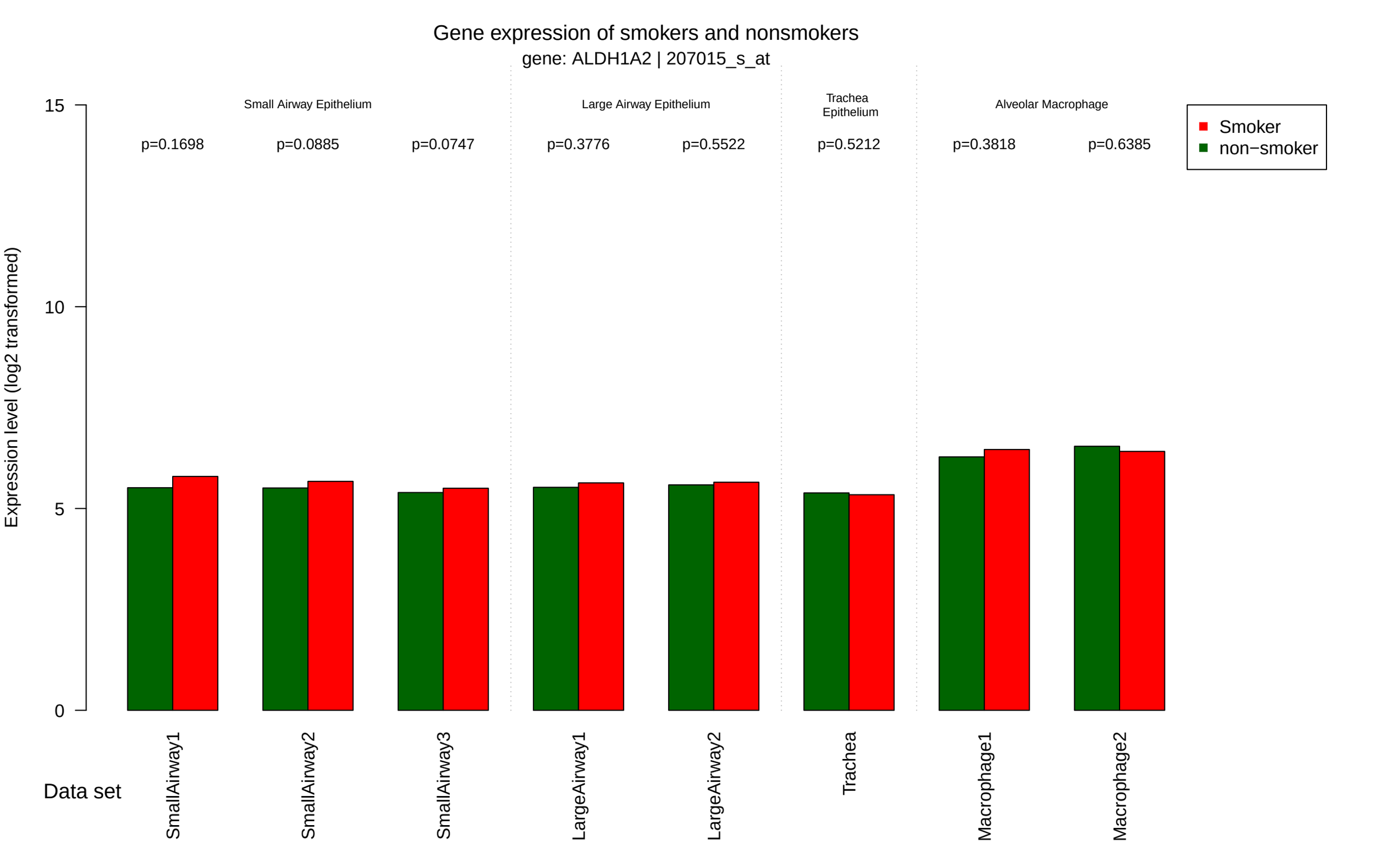
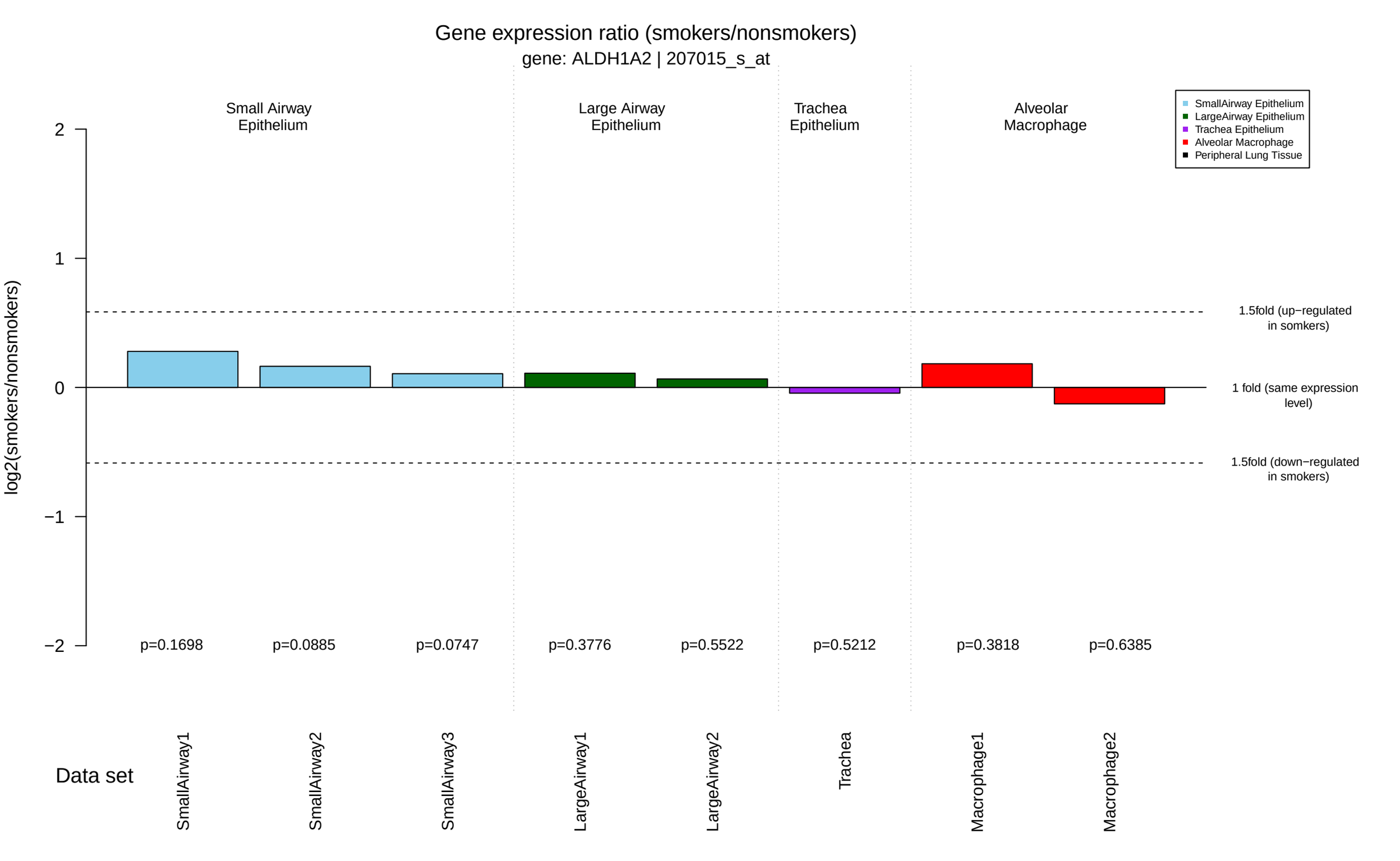
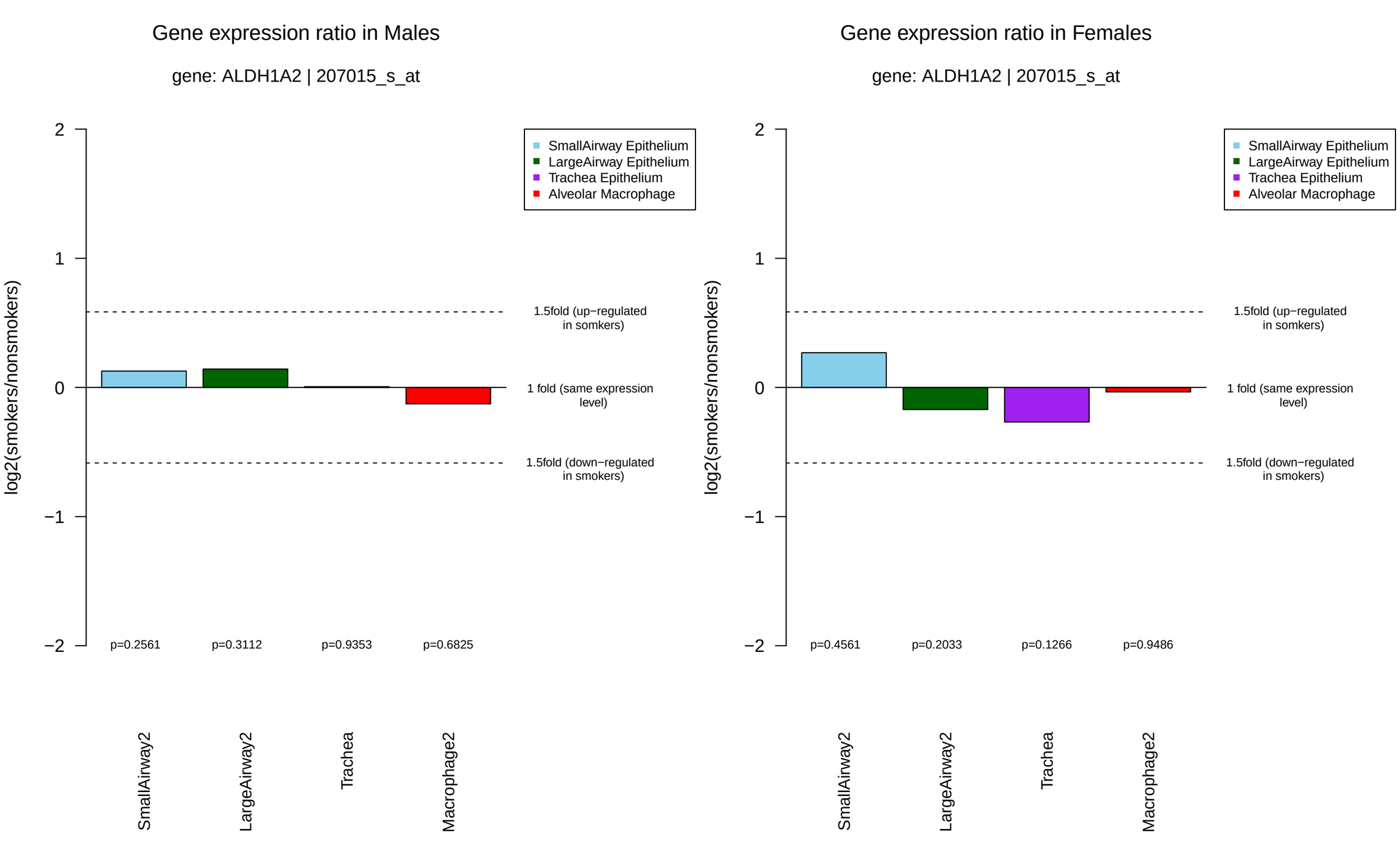
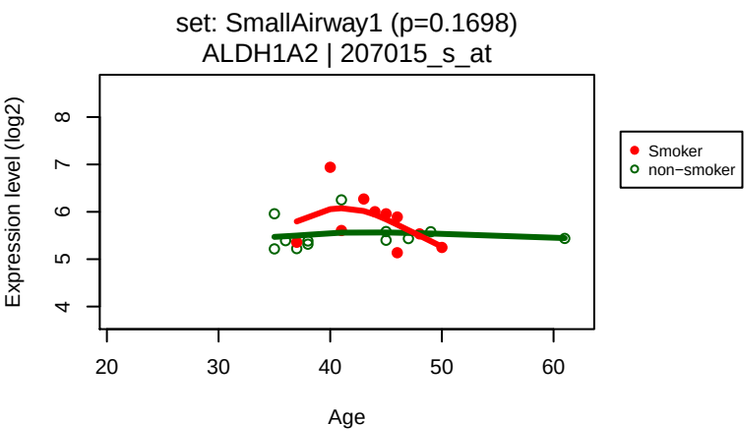
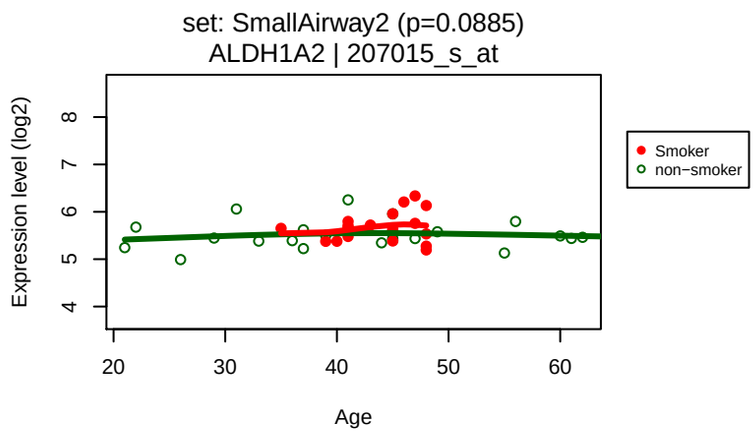
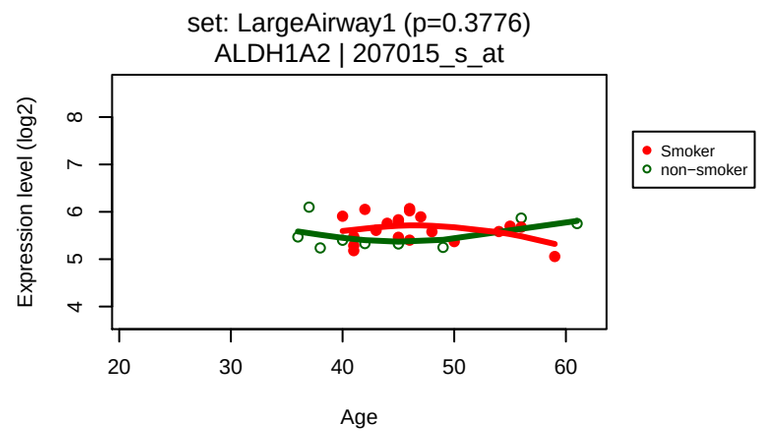

Gene expression of smokers and nonsmokers
gene: ALDH1A2 | 207015_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1698
p=0.0885
p=0.0747
p=0.3776
p=0.5522
p=0.5212
p=0.3818
p=0.6385
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

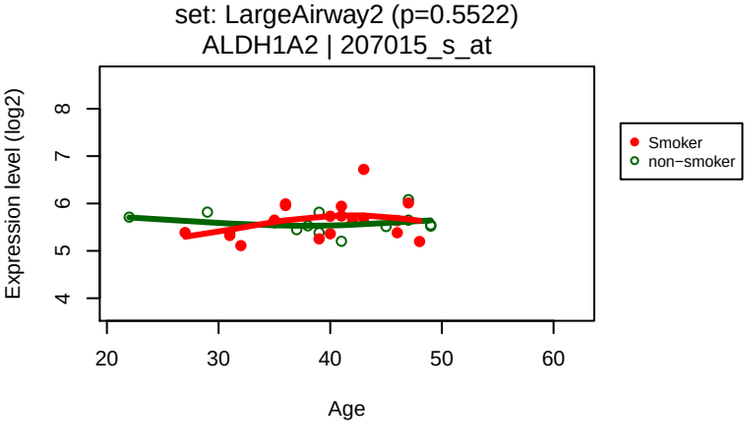

Gene expression ratio (smokers/nonsmokers)
gene: ALDH1A2 | 207015_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1698
p=0.0885
p=0.0747
p=0.3776
p=0.5522
p=0.5212
p=0.3818
p=0.6385
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

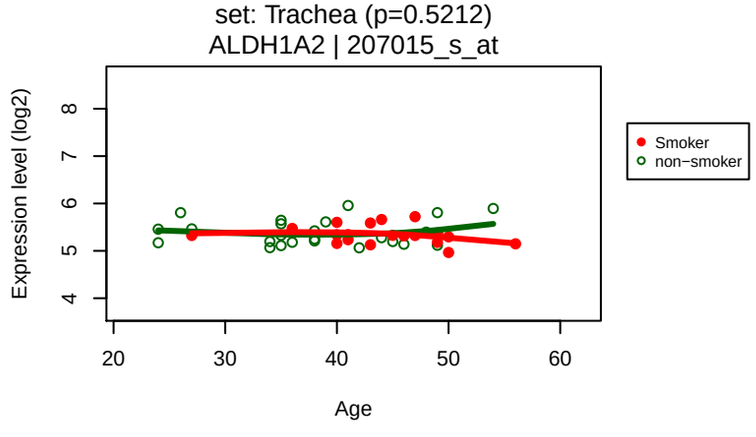

Gene expression ratio in Males
gene: ALDH1A2 | 207015_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2561
p=0.3112
p=0.9353
p=0.6825
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALDH1A2 | 207015_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4561
p=0.2033
p=0.1266
p=0.9486
SmallAirway2
LargeAirway2
Trachea
Macrophage2

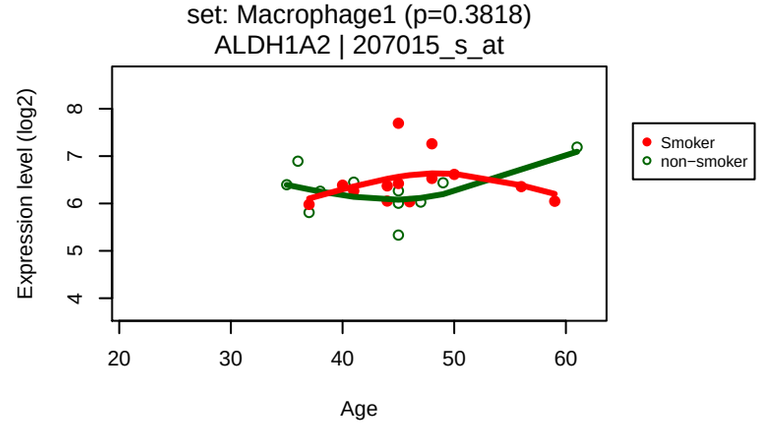

set: SmallAirway1 (p=0.1698)
ALDH1A2 | 207015_s_at
Expression level (log2)
Age
Smoker
non−smoker

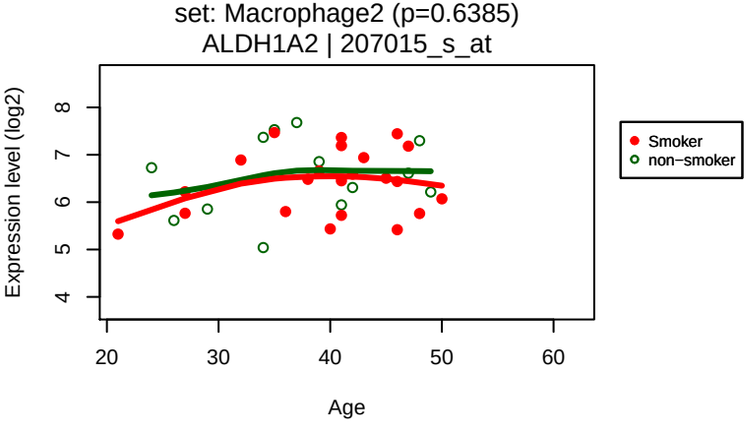

set: SmallAirway2 (p=0.0885)
ALDH1A2 | 207015_s_at
Expression level (log2)
Age
Smoker
non−smoker

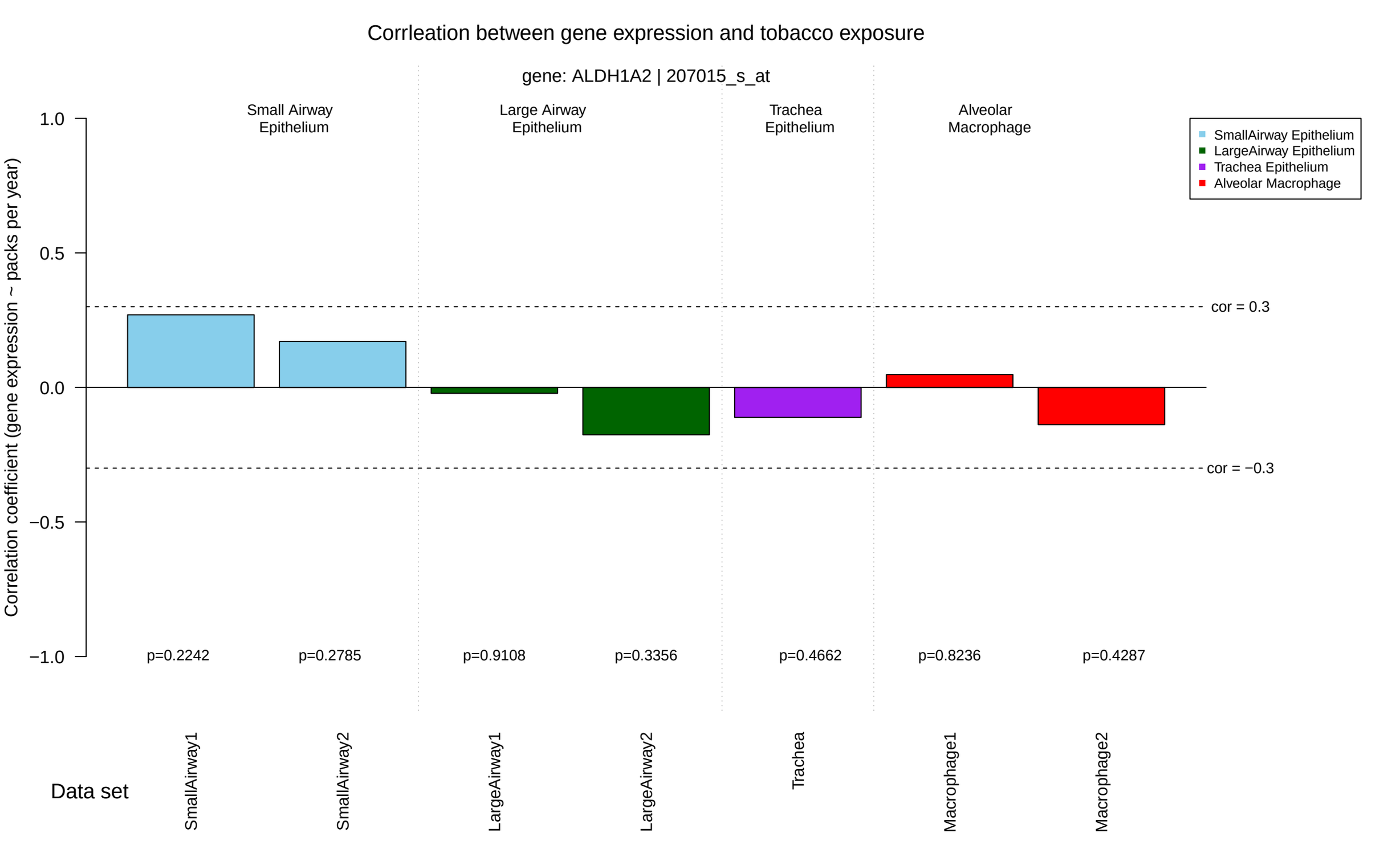

set: LargeAirway1 (p=0.3776)
ALDH1A2 | 207015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5522)
ALDH1A2 | 207015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5212)
ALDH1A2 | 207015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3818)
ALDH1A2 | 207015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6385)
ALDH1A2 | 207015_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALDH1A2 | 207015_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2242
p=0.2785
p=0.9108
p=0.3356
p=0.4662
p=0.8236
p=0.4287
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2